George P. Kontoudis

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RESEARCH INTERESTS

My research develops **learning**, **planning**, and **control** methods that enable teams of robots to safely explore unknown environments under uncertainty and with minimal communication. I work on distributed Gaussian process learning, multi-robot informative path planning, resilient distributed optimization, and quantum-accelerated learning. I also work on the design and development of adaptive robot hands and assistive devices. In parallel, I actively contribute to the OpenBionics initiative.

PROFESSIONAL & RESEARCH EXPERIENCE

Assistant Professor, Colorado School of Mines *Jan 2024–present*
Department of Mechanical Engineering

Core Faculty Member, Colorado School of Mines *Jan 2024–present*
Robotics Program

Affiliated Faculty Member, Colorado School of Mines *Aug 2025–present*
Quantum Engineering Program

Postdoctoral Research Associate, University of Maryland *Jan 2022–Dec 2023*
Motion and Teaming Lab (PI: Michael Otte)

Graduate Research Assistant, Virginia Tech *Aug 2016–Dec 2021*
Center for Marine Autonomy & Robotics (PI: Daniel J. Stilwell)
Computational Multiphysics Systems Laboratory (PI: Tomonari Furukawa)

Undergraduate Research Assistant, National Technical University of Athens *Apr 2014–Mar 2016*
Control Systems Laboratory (PI: Kostas J. Kyriakopoulos)

Founder & Research Associate, OpenBionics *Sep 2014–present*

EDUCATION

Ph.D. in Electrical Engineering, Virginia Tech *2018–2021*
Dissertation Title: "Communication-Aware, Scalable Gaussian Processes for Decentralized Exploration"

M.S. in Mechanical Engineering, Virginia Tech *2016–2018*
Thesis Title: "Adaptive, Anthropomorphic Robot Hands for Grasping and In-Hand Manipulation"

Diploma (B.S. & M.S.) in Mechanical Engineering, National Technical University of Athens *2012–2016*
Thesis Title: "Design and Development of an Underactuated, Anthropomorphic Robot Hand"

B.S. in Mechanical Engineering, University of West Attica *2005–2010*
Thesis Title: "Analysis of Compression and Air-Conditioning System in Short Takeoff and Landing Aircrafts"

TEACHING EXPERIENCE

Instructor, MEGN544/ROBO554 Robot Mechanics: Kinematics, Dynamics, and Control *Fall '24, '25, '26*
Department of Mechanical Engineering, Colorado School of Mines
Participation - Evaluation: 29 - 3.59/5.00 (F '24), 25 - 4.17/5.00 (F '25), 22 (F '26)

Instructor, MEGN315 Dynamics *Fall '25, Spring '24, '25, '26*
Department of Mechanical Engineering, Colorado School of Mines
Participation - Evaluation: 31 - 4.04/5.00 (S '24), 37 - 4.06/5.00 (S '25), 47 - 3.59/5.00 (F '25), 51 - 4.08/5.00 (S '26)

Instructor, MEGN503 Graduate Seminar *Spring '25*
Department of Mechanical Engineering (Robotics and Biomechanics Division), Colorado School of Mines
Participation - Evaluation: 12 - 4.89/5.00 (S '25)

Co-Instructor, ENAE488O/788O Introduction to Autonomous Multi-Robot Swarms *Spring '23*
Department of Aerospace Engineering, University of Maryland

Graduate Teaching Assistant, ME4005, 4006 Mechanical Engineering Lab I, II *Fall '16, Spring '17*
Department of Mechanical Engineering, Virginia Tech

SPONSORED RESEARCH PROJECTS

Total Funding: $1,850,062; Share Credit: $673,333
External (Current)

- "Gaussian Processes that Embrace Heterogeneity for Multi-Sensor Sampling Missions," *National Science Foundation, Robust Intelligence*. PIs: Michael Otte (UMD), **George P. Kontoudis** (Mines). Co-PI: Pratap Tokekar (UMD). Total award: $1,000,000; Personal share: $333,333. Duration: 09/2026–08/2029.
- "Development of Next-Generation Dynamically Adaptive Orthoses for Distal Extremity Injury," *Henry M. Jackson Foundation for the Advancement of Military Medicine*. Co-PI: Anthony Petrella. Co-I: **George P. Kontoudis**. Total award: $850,062; Personal share: $340,000. Duration: 07/2023–07/2027.

AWARDS & HONORS

NSF FRR-NRI Aspiring Robotics PI	*2023, 2024*
2022 Robotics: Science and Systems (RSS) Pioneer	*2022*
2022 MRC Postdoctoral Fellowship ($60K)	*2022*
5 × IEEE Student Travel Support for IROS, ACC, and CDC	*2015, 2019–2021*
2 × Virginia Tech GSA Travel Fund Award for Humanoids and ICORR	*2019, 2020*
NSF Student Travel Grant for WuWNet ($1K)	*2019*
NTUA Thomaideion Award for IROS	*2016*
Hackaday Prize, 2nd place among 900 projects ($10K)	*2015*
Robotdalen Innovation Award, 1st place (SEK 100K)	*2015*

INDUSTRIAL EXPERIENCE

Mechanical Engineer, Sychem S.A.	*Oct 2010–Aug 2015*
Aircraft Maintenance Engineer Trainee, Olympic Aviation	*May 2008–Jan 2010*

PUBLICATIONS

Underline denotes student advisee co-author
$^{+}$ Plus denotes undergraduate student advisee co-author
* Asterisk denotes student mentee co-author

Journal Manuscripts Under Review (J:UR)

[2] David G. Fobar, **George P. Kontoudis**, "Resilient Distributed Gaussian Processes via Trusted Edge ADMM for Decentralized Multi-Agent Systems."

[1] Mykola Mazur, **George P. Kontoudis**, Michael F. Miller, Caroline R. Turner, Katherine R. Knaus, Anne K. Silverman, Anthony Petrella, "A Personalized Dual-IMU Wearable System for Ankle Joint Kinematics Estimation During Gait."

Refereed Journal Publications (J)

[8] Josh Netter*, **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Decentralized Multi-Agent Motion Planning Using Cognitive Hierarchy and Gaussian Process Classification," *IEEE Transactions on Intelligent Vehicles*, 2026.

[7] **George P. Kontoudis**, Daniel J. Stilwell, "Multi-Agent Federated Learning using Covariance-Based Nearest Neighbor Gaussian Processes," *IEEE Transactions on Machine Learning in Communications and Networking*, 2025.

[6] **George P. Kontoudis**, Daniel J. Stilwell, "Scalable, Federated Gaussian Process Training for Decentralized Multi-Agent Systems," *IEEE Access*, 2024.

[5] Zirui Xu*, **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Online and Robust Intermittent Motion Planning in Dynamic and Changing Environments," *IEEE Transactions on Neural Networks and Learning Systems*, 2023.

[4] **George P. Kontoudis**, Stephen Krauss, Daniel J. Stilwell, "Model-Based Learning of Underwater Acoustic Communication Performance for Marine Robots," *Robotics and Autonomous Systems*, 2021.

[3] Geng Gao, Mojtaba Shahmohammadi, Lucas Gerez, **George P. Kontoudis**, Minas Liarokapis, "On Differential Mechanisms for Underactuated, Lightweight, Adaptive Prosthetic Hands," *Frontiers in Neurorobotics*, 2021.

[2] **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Kinodynamic Motion Planning with Continuous-Time Q-Learning: An Online, Model-Free, and Safe Navigation Framework," *IEEE Transactions on Neural Networks and Learning Systems*, 2019.

[1] **George P. Kontoudis**, Minas Liarokapis, Kyriakos G. Vamvoudakis, Tomonari Furukawa, "An Adaptive Actuation Mechanism for Anthropomorphic Robot Hands," *Frontiers in Robotics and AI*, 2019.

Chapters in Edited Volumes (BC)

[1] **George P. Kontoudis**, Kyriakos G. Vamvoudakis, Zirui Xu*, "RRT-QX: Real-Time Kinodynamic Motion Planning in Dynamic Environments with Continuous-Time Reinforcement Learning," in *Brain and Cognitive Intelligence: Control in Robotics*, B. Wei (Ed.), Taylor & Francis Group, CRC Press, 2022.

Refereed Conference Publications (C)

[22] Evan Pacic+, Remi Jones, Elijah Paiz, **George P. Kontoudis**, "Open-Source Body-Powered Prosthesis with a Selective Gear-Based Clutch Mechanism for Switching Grasp Types," *IEEE-RAS International Conference on Humanoid Robots **(Humanoids)***, Santa Clara, USA, 2026.

[21] Alkesh K. Srivastava*, **George P. Kontoudis**, Donald Sofge, Michael Otte, "Bayesian Networks for Path-Based Sensors: Gathering Information and Path Planning in Communication Denied Environments," *World Symposium on the Algorithmic Foundations of Robotics **(WAFR)***, Oulu, Finland, 2026.

[20] Meet Gandhi, **George P. Kontoudis**, "Distributed Quantum Gaussian Processes for Multi-Agent Systems," *International Conference on Autonomous Agents and Multiagent Systems **(AAMAS)***, Paphos, Cyprus, 2026.

[19] Sanket A. Salunkhe, **George P. Kontoudis**, "Federated Gaussian Process Learning via Pseudo-Representations for Large-Scale Multi-Robot Systems," *International Conference on Autonomous Agents and Multiagent Systems **(AAMAS)***, Paphos, Cyprus, 2026.

[18] Mykola Mazur, **George P. Kontoudis**, Anthony Petrella, "A Wearable Device for Accurate Ankle Movement Evaluation," *International Conference of the IEEE Engineering in Medicine and Biology Society **(EMBC)***, Copenhagen, Denmark, 2025.

[17] **George P. Kontoudis**, Daniel J. Stilwell, "Decentralized Federated Learning using Gaussian Processes," *IEEE International Symposium on Multi-Robot and Multi-Agent Systems **(MRS)***, Boston, USA, 2023.

[16] **George P. Kontoudis**, Michael Otte, "Adaptive Exploration-Exploitation Active Learning of Gaussian Processes," *IEEE/RSJ International Conference on Intelligent Robots and Systems **(IROS)***, Detroit, USA, 2023.

[15] **George P. Kontoudis**, Michael Otte, "Closed-Form Active Learning of Expected Variance Reduction using Gaussian Process Surrogates for Adaptive Sampling," *American Control Conference **(ACC)***, San Diego, USA, 2023.

[14] Alkesh K. Srivastava*, **George P. Kontoudis**, Donald Sofge, Michael Otte, "Distributed Multi-Robot Information Gathering using Path-Based Sensors in Entropy-Weighted Voronoi Regions," *International Symposium on Distributed Autonomous Robotic Systems **(DARS)***, Montbéliard, France, 2022. *(Also appears as a chapter in the book: Distributed Autonomous Robotic Systems, Springer Proceedings in Advanced Robotics, pp. 286–299. 2024.)*

[13] Christos N. Mavridis, **George P. Kontoudis**, John S. Baras, "Sparse Gaussian Process Regression using Progressively Growing Learning Representations," *IEEE Conference on Decision and Control **(CDC)***, Cancun, Mexico, 2022.

[12] Josh Netter*, **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Bounded Rational RRT-QX: Multi-Agent Motion Planning in Dynamic Human-Like Environments Using Cognitive Hierarchy and Q-Learning," *IEEE Conference on Decision and Control **(CDC)***, Austin, USA, 2021.

[11] **George P. Kontoudis**, Daniel J. Stilwell, "Decentralized Nested Gaussian Processes for Multi-Robot Systems," *IEEE International Conference on Robotics and Automation **(ICRA)***, Xi'an, China, 2021.

[10] Minas Liarokapis, **George P. Kontoudis**, "Teaching Robotic and Biomechatronic Concepts with a Gripper Design Project and a Grasping and Manipulation Competition," *IEEE International Conference on Robotics and Automation (**ICRA**)*, Xi'an, China, 2021.

[9] **George P. Kontoudis**, Daniel J. Stilwell, "Prediction of Acoustic Communication Performance in Marine Robots Using Model-Based Kriging," *American Control Conference (**ACC**)*, New Orleans, USA, 2021.

[8] Gal Gorjup, **George P. Kontoudis**, Anany Dwivedi, Geng Gao, Saori Matsunaga, Toshisada Mariyama, Bruce MacDonald, Minas Liarokapis "Combining Programming by Demonstration with Path Optimization and Local Replanning to Facilitate the Execution of Assembly Tasks ," *IEEE International Conference on Systems, Man and Cybernetics (**SMC**)*, Toronto, Canada, 2020.

[7] **George P. Kontoudis**, Zirui Xu*, Kyriakos G. Vamvoudakis, "Online, Model-Free Motion Planning in Dynamic Environments: An Intermittent, Finite Horizon Approach with Continuous-Time Q-Learning," *American Control Conference (**ACC**)*, Denver, USA, 2020.

[6] **George P. Kontoudis**, Daniel J. Stilwell, "A Comparison of Kriging and Cokriging for Estimation of Underwater Acoustic Communication Performance," *ACM International Conference on Underwater Networks and Systems (**WuWNet**)*, Atlanta, USA, 2019.

[5] **George P. Kontoudis**, Minas Liarokapis, Kyriakos G. Vamvoudakis, "An Adaptive, Humanlike Robot Hand with Selective Interdigitation: Towards Robust Grasping and Dexterous, In-Hand Manipulation," *IEEE-RAS International Conference on Humanoid Robots (**Humanoids**)*, Toronto, Canada, 2019.

[4] **George P. Kontoudis**, Minas Liarokapis, Kyriakos G. Vamvoudakis, "A Compliant, Underactuated Finger for Anthropomorphic Hands," *IEEE/RAS-EMBS Inter. Conference on Rehabilitation Robotics (**ICORR**)*, Toronto, Canada, 2019.

[3] **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Robust Kinodynamic Motion Planning using Model-Free Game-Theoretic Learning," *American Control Conference (**ACC**)*, Philadelphia, USA, 2019.

[2] Kyriakos D. Tsoukalas, **George P. Kontoudis**, Kyriakos G. Vamvoudakis, "Active-Bayesian Learning for Cooperation Connectivity in Dynamic Cyber-Physical-Human Systems," *IEEE Symposium on Adaptive Dynamic Programming and Reinforcement Learning (**ADPRL**)*, Honolulu, USA, 2017.

[1] **George P. Kontoudis**, Minas Liarokapis, Agisilaos G. Zisimatos, Christoforos I. Mavrogiannis, Kostas J. Kyriakopoulos, "Open-Source, Anthropomorphic, Underactuated Robot Hands with a Selectively Lockable Differential Mechanism: Towards Affordable Prostheses," *IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)*, Hamburg, Germany, 2015.

Refereed Workshop Publications (W)

[3] Sanket A. Salunkhe, **George P. Kontoudis**, "Multi-Agent Informative Sampling using Distributed Gaussian Processes," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Workshop on Intelligent Information Gathering for Single and Multi-Robot Systems*, Pittsburgh, USA, 2026.

[2] Alkesh K. Srivastava*, **George P. Kontoudis**, Donald Sofge, Michael Otte, "Path-Based Sensors: Will the Knowledge of Correlation in Random Variables Accelerate Information Gathering?" *IEEE International Conference on Robotics and Automation (ICRA), Workshop on Communication Challenges in Multi-Robot Systems: Perception, Coordination, and Learning*, London, UK, June 2023.

[1] **George P. Kontoudis**, "Scalable Multi-Robot Active Exploration," *Robotics: Science and Systems (RSS), Pioneers Workshop*, New York City, USA, June 2022.

Refereed Abstract Publications (A)

[1] Mykola Mazur, **George P. Kontoudis**, Anthony Petrella, "A Wireless Sensor Platform for Distal Extremity Movement Evaluation," *Orthopedic Research Society (ORS) Annual Meeting*, Phoenix, USA, February 2025.

Theses (T)

[3] **George P. Kontoudis**, "Communication-Aware, Scalable Gaussian Processes for Decentralized Exploration," *Doctoral Dissertation, Virginia Tech*, USA, December 2021.

[2] **George P. Kontoudis**, "Adaptive, Anthropomorphic Robot Hands for Grasping and In-Hand Manipulation," *Master's Thesis, Virginia Tech*, USA, December 2018.

[1] **George P. Kontoudis**, "Design and Development of an Underactuated, Anthropomorphic Robot Hand," *Diploma Thesis, National Technical University of Athens*, March 2016. *(in Greek)*

Technical Reports (TR)

[3] **George P. Kontoudis**, Daniel J. Stilwell, "Fully Decentralized, Scalable Gaussian Processes for Multi-Agent Federated Learning," *arXiv preprint, arXiv:2203.02865*, 2022.

[2] **George P. Kontoudis**, Minas Liarokapis, Agisilaos G. Zisimatos, Christoforos I. Mavrogiannis, Kostas J. Kyriakopoulos, "How to Create Affordable, Anthropomorphic, Light-Weight Prosthetic Hands," *Control Systems Lab, National Technical University of Athens,* Athens, Greece, October 2015.

[1] Agisilaos G. Zisimatos, Minas Liarokapis, Christoforos I. Mavrogiannis, **George P. Kontoudis**, Kostas J. Kyriakopoulos, "How to Create Affordable, Modular, Light-Weight, Underactuated, Compliant Robot Hand," *Control Systems Lab, National Technical University of Athens,* Athens, Greece, January 2015.

TALKS & PRESENTATIONS

- "Tips for Writing Technical Papers," *Summer Undergraduate Research (**SURF**) Lecture, Colorado School of Mines*, Golden, USA, 2026. **[Oral Presentation]**
- "Distributed Quantum Gaussian Processes for Multi-Agent Systems," *International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**)*, Paphos, Cyprus, 2026. **[Oral Presentation]**
- "Federated Gaussian Process Learning via Pseudo-Representations for Large-Scale Multi-Robot Systems," *International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**)*, Paphos, Cyprus, 2026. **[Oral Presentation]**
- "Learning to Act Together: Distributed Gaussian Processes for Scalable Multi-Robot Planning and Control," *Cyprus University of Technology*, Limassol, Cyprus, 2026. **[Invited Talk]**
- "Learning Underwater Communication Maps for Large-Scale Multi-Robot Systems," *AOPE Seminar, Woods Hole Oceanographic Institution (WHOI)*, Woods Hole, USA, 2026. **[Invited Talk]**
- "From Information to Dynamics: Optimal Kinodynamic and Informative Path Planning," *Robotics Seminar, CU Boulder*, Boulder, USA, 2025. **[Invited Talk]**
- "Adaptive Informative Path Planning under Uncertainty via Gaussian Processes," *American Control Conference (**ACC**), Workshop: Data-Driven and Risk-Aware Control for Safety-Critical Autonomous Systems*, Denver, USA, 2025. **[Invited Talk]**
- "Affordable Adaptive Robot Hands for Grasping and Manipulation," *NSF RET, Colorado School of Mines*, Golden, USA, 2025. **[Oral Presentation]**
- "Decentralized Informative Path Planning with Gaussian Processes," *NSF FRR-NRI PI Meeting*, Baltimore, USA, 2024. **[Poster Presentation]**
- "Optimal Kinodynamic Motion Planning and Informative Path Planning," *Computer Science and Robotics Seminar, Colorado School of Mines*, Golden, USA, 2024. **[Oral Presentation]**
- "Decentralized Federated Learning using Gaussian Processes," *IEEE International Symposium on Multi-Robot and Multi-Agent Systems (**MRS**)*, Boston, USA, 2023. **[Oral Presentation]**
- "Adaptive Exploration-Exploitation Active Learning of Gaussian Processes," *IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)*, Detroit, USA, 2023. **[Oral & Poster Presentation]**
- "Closed-Form Active Learning Using Expected Variance Reduction of Gaussian Process Surrogates for Adaptive Sampling," *American Control Conference (**ACC**)*, San Diego, USA, 2023. **[Oral Presentation]**
- "Decentralized Gaussian Process Learning for Exploration with Multi-Robot Systems," *Maryland Robotics Center Research Symposium*, College Park, USA, May 2023. **[Oral Presentation - Invited Talk]**
- "Scalable Exploration-Exploitation Active Learning of Gaussian Process Surrogates for Adaptive Sampling," *Maryland Robotics Center Research Symposium*, College Park, USA, May 2023. **[Poster Presentation]**
- "Multi-Robot Adaptive Sampling using Decentralized Gaussian Processes," *International Symposium on Distributed Autonomous Robotic Systems (**DARS**)*, Montbéliard, France, November 2022. **[Poster Presentation]**
- "Scalable Multi-Robot Active Exploration," *Robotics: Science and Systems (**RSS**), **Pioneers Workshop***, New York City, USA, June 2022. **[Spotlight Presentation]**
- "Communication-Aware, Scalable Gaussian Processes for Decentralized Exploration," *Bradley Department of Electrical and Computer Engineering, Virginia Tech*, Blacksburg, USA, December 2021. **[PhD Defense]**
- "Decentralized Nested Gaussian Processes for Multi-Robot Systems," *IEEE International Conference on Robotics and Automation (**ICRA**)*, Xi'an, China, 2021. **[Virtual Presentation]**

- "Online, Model-Free Motion Planning in Dynamic Environments: An Intermittent, Finite Horizon Approach with Continuous-Time Q-Learning," *American Control Conference **(ACC)***, Denver, USA, 2020. **[Rapid-Interactive Presentation]**
- "A Comparison of Kriging and Cokriging for Estimation of Underwater Acoustic Communication Performance," *ACM International Conference on Underwater Networks and Systems **(WuWNet)***, Atlanta, USA, 2019. **[Oral Presentation]**
- "An Adaptive, Humanlike Robot Hand with Selective Interdigitation: Towards Robust Grasping and Dexterous, In-Hand Manipulation," *IEEE-RAS Intern. Conference on Humanoid Robots **(Humanoids)***, Toronto, Canada, 2019. **[Poster Presentation]**
- "Robust Kinodynamic Motion Planning using Model-Free Game-Theoretic Learning," *American Control Conference **(ACC)***, Philadelphia, USA, 2019. **[Oral Presentation]**
- "A Compliant, Underactuated Finger for Anthropomorphic Hands," *IEEE/RAS-EMBS International Conference on Rehabilitation Robotics **(ICORR)***, Toronto, Canada, 2019. **[Poster Presentation]**
- "Adaptive, Anthropomorphic Robot Hands for Grasping and In-Hand Manipulation," *Department of Mechanical Engineering, Virginia Tech*, Blacksburg, USA, December 2018. **[Master's Defense]**
- "Evaluation Strategies of Adaptive, Anthropomorphic Robot Hands for Dexterous In-Hand Manipulation: Early Results," *National Institute of Standards and Technology **(NIST)***, USA, 2018. **[Invited Talk]**
- "Open-Source, Anthropomorphic, Underactuated Robot Hands with a Selectively Lockable Differential Mechanism: Towards Affordable Prostheses," *IEEE/RSJ International Conference on Intelligent Robots and Systems **(IROS)***, Hamburg, Germany, 2015. **[Oral Presentation]**

SERVICE ACTIVITIES

Editorial Service

- Associate Editor, IEEE Robotics and Automation Letters (RAL) — *2025–present*
- Associate Editor, IEEE International Conference on Robotics and Automation (ICRA) — *2025*
- Area Chair, Robotics: Science and Systems (RSS), Pioneers — *2025, 2026*

Grant Proposal Review

- National Science Foundation (NSF) Reviewer — *2024, 2026*

Conference Committees

- Program Committee Member, Australasian Conference on Robotics and Automation (ACRA) — *2024*
- General Chair, Robotics: Science and Systems (RSS), Pioneers Workshop — *2023*
- Program Committee Member, Conference on Artificial Intelligence (AAAI), Student Abstract and Poster — *2023, 2024*
- Online Platform Chair, Conference on Robot Learning (CoRL) — *2022*

Peer Review

- **Journals**: IEEE Transactions on Neural Networks and Learning Systems (TNNLS) · IEEE Transactions on Robotics (TRO) · IEEE Transactions on Automatic Control (TAC) · Automatica · IEEE Transactions on Automation Science and Engineering (TASE) · International Journal of Robotics Research (IJRR) · IEEE Robotics and Automation Letters (RAL) · IEEE Access (ACCESS) · IEEE Control Systems Letters (CSL) · IEEE Robotics & Automation Magazine (RAM) · Autonomous Robots (AURO) · IEEE Journal of Oceanic Engineering (JOE) · IEEE Open Journal of Signal Processing (OJSP) · IEEE Transactions on Aerospace and Electronic Systems(TAES) · IEEE Transactions on Systems, Man and Cybernetics: Systems (TSMCA) · IEEE Transactions on Cybernetics · IEEE Computational Intelligence Magazine (CIM) · Frontiers in Robotics and AI (FROBAI) · Frontiers in Artificial Intelligence (FAI) · Journal of Optimization Theory and Applications (JOTA) · International Journal of Advanced Robotic Systems (IJARS)
- **Conferences**: IEEE International Conference on Robotics and Automation (ICRA) · IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) · Robotics: Science and Systems (RSS) · American Control Conference (ACC) · IEEE Conference on Decision and Control (CDC) · International Symposium on Distributed Autonomous Robotic Systems (DARS) · IEEE Conference on Control Technology and Applications (CCTA) · International Conference on Ubiquitous Robots (UR) · IEEE International Conference on Automation Science and Engineering (CASE) · IEEE-RAS International Conference on Humanoid Robots (Humanoids) · European Control Conference (ECC) · IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob) · Mediterranean Conference on Control and Automation (MED)

University Service

- Graduate Admission Committee Member *2024–present*
  Mechanical Engineering Department, *Colorado School of Mines*
- Faculty Grievance Committee Member *2026–present*
  Colorado School of Mines
- Graduate Admissions Committee Member *2026*
  Robotics Program, *Colorado School of Mines*
- **Judge**: Mines Graduate Research and Discovery Symposium (GRADS) - S '24, S '25 · Mines Capstone Design Senior Showcase - S '24, S '25 · Mines Undergraduate Research Symposium - F '24, S '25, S '26

RESEARCH ADVISING

PhD Students

- Aaron Fanganello, Robotics, Colorado School of Mines *2026–present*
  Awards: 2026 SMART Scholar
- Dave Fobar, Robotics, Colorado School of Mines *2025–present*
  Publications: J:UR-[2]
- Meet Gandhi, Robotics, Colorado School of Mines *2025–present*
  Publications: C-[20]
  Awards: 2025 Mechanical Engineering Fellowship, 2026 AAMAS Doctoral Consortium, 2026 AAMAS Student Scholarship
- Mykola Mazur (co-advised with Anthony Petrella), Mechanical Engineering, Colorado School of Mines *2024–present*
  Publications: A-[1], C-[18], J:UR-[1]
- Sanket Salunkhe, Robotics, Colorado School of Mines *2024–present*
  Publications: C-[19], W-[3]
  Awards: 2026 AAMAS Student Scholarship
- Joshua Netter (*mentored with Kyriakos G. Vamvoudakis*), Georgia Institute of Technology *2020–2023*
  Publications: C-[12], J-[8]
  Next Position: Aerospace Algorithms Engineer, AST SpaceMobile

Master's Students

- Sam Gregerson, Colorado School of Mines *2026–present*
- Nick Wyman, Colorado School of Mines *2026–present*
- Leo Paparoni (co-advised with Anthony Petrella), Colorado School of Mines *2026–present*
- Olivia Franklin, Colorado School of Mines *2026–present*
- Remi Jones, Colorado School of Mines *2026–present*
  Publications: C-[22]
- Fahad Asat, Colorado School of Mines *2026–present*
- Elijah Paiz, Colorado School of Mines *2025–2026*
  Publications: C-[22]
  Next Position: Strength Analyst, United Launch Alliance (ULA)
- Jude Ortiz, Colorado School of Mines *2025–2026*
  Next Position: Additive Engineer, Big Metal Additive
- Alkesh K. Srivastava (*mentored with Michael Otte*), University of Maryland *2022–2023*
  Publications: C-[14], W-[2], C-[21]
  Next Position: PhD student, Temple University
- Zirui Xu (*mentored with Kyriakos G. Vamvoudakis*), Georgia Institute of Technology *2018–2020*
  Publications: C-[7], BC-[1], J-[5]
  Next Position: PhD student, University of Michigan

Undergraduate Students

- Graden Gentry, Colorado School of Mines *2026–present*
- Analissia Wise, Colorado School of Mines *2026–present*
- Audrey Smith, Colorado School of Mines *2025–present*
  Awards: 2026 FIRST Cohort, 2026 SURF Scholar, and 2027 MURF Scholar

- Matthew Smith, Colorado School of Mines *2025–present*
- Arael Anaya, Colorado School of Mines *2025–present*
  Awards: 2025 IEEE CASE Travel Grant, 2025 SURF Scholar, and 2026 MURF Scholar
- Evan Pacic, Colorado School of Mines *2024–present*
  Awards: 2025 FIRST Cohort, 2026 MURF and SURF Scholar
  Publications: C-[22]
- Olivia Franklin, Colorado School of Mines *2025–2026*
  Next Position: MS student, Colorado School of Mines
- Remi Jones, Colorado School of Mines *2024–2026*
  Awards: 2nd poster place 2024 Fall Undergraduate Research Symposium, 2026 Undergraduate Research Scholar Distinction
  Next Position: MS student, Colorado School of Mines
- Jude Ortiz, Colorado School of Mines *2025*
  Next Position: MS student, Colorado School of Mines
- Elijah Paiz, Harvey Scholar, Colorado School of Mines *2024–2025*
  Awards: 2nd poster place 2024 Fall Undergraduate Research Symposium
  Next Position: MS student, Colorado School of Mines
- Ross Hodson, Colorado School of Mines
  Awards: 2024 SURF Scholar and 2025 MURF Scholar *2024–2025*
  Next Position: MS student, Oregon State University

Visiting Students

- Ethan Brown, FIRE Scholar '25, Brigham Young University *2025*
  Next Position: MS student, Brigham Young University

PhD Students - Dissertation Committee Member

- Thong Quoc Huynh, Robotics, Colorado School of Mines *2026–present*
- Eugene Hamzezadeh, Robotics, Colorado School of Mines *2025–present*
- Matthew Stanley, Mechanical Engineering, Colorado School of Mines *2024–present*
- Bailey Hopkins, Robotics, Colorado School of Mines *2026*
- Seth Stewart, Robotics, Colorado School of Mines *2025*

MS Students - Thesis Committee Member

- Yee Shen Teoh, Computer Science, Colorado School of Mines *2024–2025*

PhD Students - Qualifying Exam Committee Member

- Will Feldman, Mechanical Engineering, Colorado School of Mines *Fall 2026*
- Linda Knudson, Mechanical Engineering, Colorado School of Mines *Spring 2026*
- Jiuyi Xu, Robotics, Colorado School of Mines *Spring 2026*
- Haozhixiang Li, Mechanical Engineering, Colorado School of Mines *Fall 2025*
- Kazi Rifat Bin Rafiq, Mechanical Engineering, Colorado School of Mines *Fall 2024, Spring 2025*
- James Verheyden, Mechanical Engineering, Colorado School of Mines *Fall 2024*
- Eugene Hamzezadeh, Robotics, Colorado School of Mines *Spring 2024*
- Seth Stewart, Robotics, Colorado School of Mines *Spring 2024*